

Project 1

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Data Science and Applications

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a) Who works more hours, men or women?

Men do, by an average of 6.02 hours per week (42.43 hours for men vs. 36.41 for women). This matches the sex_male coefficient from Model 1 (hours_per_week regressed on sex alone), confirming that the group means and the regression agree.

b) If education_num is added as a control variable, does the trend in hours worked by men vs. women remain the same?

Yes, essentially. Adding education_num moves the sex_male coefficient only slightly, from 6.02 to 5.97 hours (Model 2), so the gap between men and women is not explained by differences in education level.

c) Is education_num significant?

Yes. It is the only variable among age, education_num, and hours_per_week whose pairwise correlation exceeds |0.1| (r = 0.148 with hours_per_week), and a Pearson test on that pair returns p = 4.24 x 10^-159, far below any common threshold. This is consistent with the stable ~0.7-hour increase in hours_per_week per year of schooling seen in Model 2.

d) Comparing the model with gross_income_group added (3 variables) to the 2 and 1 variable models, which statistic would you use to decide the best model?

R-squared rises with each added variable (0.053, 0.074, 0.094 for Models 1, 2, and 3), but R-squared always increases whenever a predictor is added, even a weak one, so it is not a fair way to compare models with different numbers of variables on its own. Adjusted R-squared, or an F-test comparing the nested models, is the more appropriate statistic, since it checks whether the improvement

in fit is large enough to be real. Given the large gross_income_group coefficient (4.52 extra hours/week for incomes above $50K) and the already-confirmed significance of education_num, the 3-variable model is preferred.

e) Do any of the variables appear to be correlated? How did you make your assessment?

Using a Pearson correlation matrix across age, education_num, and hours_per_week, all three pairwise correlations are weak: education_num and hours_per_week are strongest at r = 0.148, hours_per_week and age at r = 0.069, and age and education_num at r = 0.037. None show a strong relationship, though education_num and hours_per_week are the closest to a real association.

f) Statistically test any pair with |r| > 0.1 for its difference from 0 and report the result. Are the direction and significance as expected?

Only education_num vs. hours_per_week exceeds |r| = 0.1. The Pearson test gives r = 0.148 and p = 4.24 x 10^-159, so the correlation is statistically significant. The positive direction (more schooling, more hours worked) matches expectation, but the magnitude is weaker than one might assume given how central education is often treated as a driver of labor supply.

g) How does the correlation between education_num and age compare between male and female participants? Is this expected?

For women, r = -0.018 with p = 0.063 (not significant). For men, r = 0.061 with p = 4.02 x 10^-19 (significant despite being small, due to the large male sample, n = 21,790). Among men there is a very weak but statistically real positive relationship between age and education; among women there is none.

This is a small but real difference, possibly reflecting generational differences in education levels between men and women, though both effect sizes are too small to matter in practice.

h) Compute the covariance matrix for education_num and hours_per_week. What conclusions can you draw from it?

Var(education_num) = 6.62, Var(hours_per_week) = 152.46, and Cov(education_num, hours_per_week) = 4.71. The positive covariance confirms the same positive direction seen in the correlation, but since covariance is not standardized, its size alone does not say much about strength. That is what the correlation coefficient (r = 0.148) is for; the covariance mainly confirms the sign already established by the correlation.